

AGL Rwanda Data Dictionary

Verified Structure Reference — built directly from the generated files, not assumed

This replaces the data-dictionary assumptions used in the earlier orchestration guide. Instead of inferring column types from CSV text, every table below was read with `pandas` / `pyarrow` straight from the `.parquet` output (the one format that carries its own real schema), and every foreign-key relationship called out in Part 8 was tested by actually computing set overlaps between the ID columns — not assumed from naming.

WHAT WAS WRONG BEFORE

Two things in the prior document turned out to be incorrect once checked against the real files: (1) it assumed all tables were exported in all 4 formats (CSV/JSON/Parquet/XML) — `customers_master`, `shippers_master` and `consignees_master` only exist as CSV/JSON; and (2) row-count estimates came from counting raw text lines, which over-counts any table whose `address` field contains an embedded newline (e.g. `customers_master` looked like ~400 rows by line count but is actually 200 records). Every count in this document is a real `len(df)`, not a line count.

Folder	Tables	Total rows (verified)
agl_data_output	4 (1 fact + 3 master)	1,450
agl_fleet_data_output	5	1,073
agl_driver_data_output	6	34,807
agl_coldchain_data_output	6	10,307
agl_delivery_data_output	5	13,049
agl_external_data_output	5	9,632
Total	31 tables	70,318 rows

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PART 1

agl_data_output — Shipment & Master Data

agl_shipment_data

1,000 rows · formats: CSV, JSON, Parquet, XML · grain: one row per shipment · primary key: `shipment_id` (format `SHIP_#####`)

Column	Type	Nulls	Sample
shipment_id	string	0	SHIP_00000001
customer_id	string	0	CUST_000144
customer_name	string	0	Hernandez and Sons
shipper_id	string	0	SHIP_000084
shipper_name	string	0	Braun and Sons
consignee_id	string	0	CONS_000006
consignee_name	string	0	Parker-Cole
billing_account	string	0	GB73GSWH26808971831622
weight_kg	double	0	6828.19
volume_m3	double	0	97.72
piece_count	integer	0	26
hs_code	string	0	9401.61
cargo_classification	string	0	Perishable
cargo_description	string	0	Indicate guess society.
container_type	string	0	LCL
freight_charges	double	0	13377.74
customs_duties	double	0	1763.70
tax_amount	double	0	1789.79
total_cost	double	0	16931.23
currency	string	0	GBP
invoice_status	string	0	Draft
invoice_number	string	0	INV_SHIP_00000001
invoice_date	timestamp	0	2026-02-26 08:40:47
order_date	timestamp	0	2026-02-23 08:40:47
dispatch_date	timestamp	0	2026-02-25 08:40:47
delivery_deadline	timestamp	0	2026-04-03 08:40:47
actual_delivery_date	timestamp	222 (22%)	2026-07-23 23:01:30 — null while still in transit
guaranteed_transit_days	integer	0	24
penalty_threshold_days	integer	0	5

penalty_rate	double	0	0.04
sla_compliance	string	0	Compliant
delay_reason	string	0	Traffic
status	string	0	In Transit
priority	string	0	Low
created_at	timestamp	0	2026-02-23 08:40:47
updated_at	timestamp	0	2026-07-04 08:46:58
notes	string	0	Successful claim condition to receive seek political stage.

customers_master

200 rows · formats: **CSV, JSON only** (no Parquet/XML) · primary key: `customer_id` (`CUST_#####`)

Column	Type	Nulls	Sample
customer_id	string	0	CUST_000000
customer_name	string	0	Lowe-Flores
customer_type	string	0	Freight Forwarder
credit_limit	double	0	64552.22
payment_terms	string	0	Net 90
address	string (multi-line)	0	5629 Patricia Curve Suite 716 / North Jenniferburgh, MT 93878
contact_email	string	0	austiningram@example.com
contact_phone	string	0	001-336-680-3752x398

shippers_master

100 rows · formats: **CSV, JSON only** · primary key: `shipper_id` (`SHIP_#####` , 6-digit — see Part 8 naming-collision note)

Column	Type	Nulls	Sample
shipper_id	string	0	SHIP_000000
shipper_name	string	0	Sanchez-Wilson
address	string (multi-line)	0	327 Randy River Suite 729 / Port Martha, CA 53914
contact_person	string	0	Tina Foster
contact_email	string	0	shannon25@example.net
contact_phone	string	0	333.415.8169x26015

consignees_master

150 rows · formats: **CSV, JSON only** · primary key: `consignee_id` (`CONS_#####`)

Column	Type	Nulls	Sample
consignee_id	string	0	CONS_000000
consignee_name	string	0	Payne, Trevino and Garcia
address	string (multi-line)	0	331 Valerie Turnpike Apt. 082 / West Amandashire, VI 26339
contact_person	string	0	Thomas Melendez

contact_email	string	0	douglas57@example.com
contact_phone	string	0	860.723.5690x481

VERIFIED GOOD

`agl_shipment_data.customer_id/shipper_id/consignee_id` all join cleanly to their master tables — 199/199, 100/100 and 150/150 of the shipment file's unique IDs were found in the respective master table. This part of the dataset is consistent.

PART 2

agl_fleet_data_output — Fleet & Telematics

All 5 tables exported in CSV, JSON, Parquet and XML.

vehicles

50 rows · primary key: `vehicle_id` (`VEH_000001` , 6-digit)

Column	Type	Nulls	Sample
vehicle_id	string	0	VEH_000001
license_plate	string	0	865 GMU
model	string	0	Mercedes Actros
year	integer	0	2015
fuel_capacity_l	integer	0	900
avg_consumption_l_per_100km	integer	0	30
current_fuel_level	double	0	667.4
odometer_reading_km	double	0	126221.47
truck_type	string	0	Heavy
max_load_tonnes	double	0	32.09
status	string	0	Active
assigned_driver	string — full name, not an ID	0	Paige Coleman
assigned_corridor	string	0	Kigali-Gisenyi
insurance_expiry	timestamp	0	2028-05-26 18:39:45
last_inspection_date	timestamp	0	2026-02-12 18:27:15
next_inspection_date	timestamp	0	2026-07-24 21:18:27

trips

500 rows · primary key: `trip_id` (`TRIP_#####`) · FK `vehicle_id`  matches `vehicles` (50/50)

Column	Type	Nulls	Sample
trip_id	string	0	TRIP_00000001
vehicle_id	string	0	VEH_000035
corridor_name	string (3 distinct values)	0	Kigali-Gatuna
start_location	string	0	Kigali
end_location	string	0	Gatuna
planned_distance_km	integer	0	340
actual_distance_km	double	0	330.84
planned_duration_hours	integer	0	7

actual_duration_hours	double	0	9.54
start_time	timestamp	0	2026-04-17 06:46:49
end_time	timestamp	0	2026-04-17 16:19:30
fuel_consumed_liters	double	0	105.71
avg_speed_kmh	double	0	34.66
max_speed_kmh	double	0	66.25
idle_time_minutes	double	0	58.13
toll_booths_passed	integer	0	3
border_crossings	integer	0	0
delay_minutes	integer	0	55
delay_reason	string	0	Weather
cargo_weight_tonnes	double	0	20.07
cargo_type	string	0	Hazardous
temperature_c	double	0	23.73
humidity_percent	double	0	89.05
driver_name	string — full name, not an ID	0	Amy Heath
status	string	0	Completed

gps_tracking

201 rows · FK `vehicle_id`  matches `vehicles` · FK `trip_id`  **zero overlap** with `trips.trip_id` (199 unique GPS trip_ids vs 500 trip trip_ids, no match — see Part 8)

Column	Type	Nulls	Sample
vehicle_id	string	0	VEH_000001
trip_id	string	0	TRIP_004835
corridor_name	string	0	Kigali-Gisenyi
latitude	double	0	-1.943415
longitude	double	0	30.061409
altitude_m	double	0	1466.7
speed_kmh	double	0	33.39
heading_deg	double	0	270.58
timestamp	timestamp	0	2026-07-15 06:08:43.126295
near_toll_booth	boolean	0	False
toll_booth_name	string	193 (96%)	set only when near_toll_booth=True
near_border_post	boolean	0	False
border_post_name	string	180 (90%)	set only when near_border_post=True
geofence_entry	boolean	0	False
geofence_exit	boolean	0	False
distance_traveled_km	double	0	0.0

telematics

182 rows · FK vehicle_id 

Column	Type	Nulls	Sample
reading_id	string	0	TEL_C5A8D954
vehicle_id	string	0	VEH_000001
timestamp	timestamp	0	2026-07-12 19:08:43.158496
fuel_level_l	double	0	509.38
fuel_consumption_rate_l_per_100km	double	0	33.17
odometer_reading_km	double	0	126697.99
engine_rpm	integer	0	1562
coolant_temp_c	integer	0	79
fuel_pressure_bar	double	0	2.61
battery_voltage	double	0	12.82
error_codes	string	0 (value "None" as text)	None
engine_load_percent	double	0	11.81
transmission_temp_c	double	0	110.18
brake_pressure_bar	double	0	7.7
tire_pressure_front_kpa	double	0	608.82
tire_pressure_rear_kpa	double	0	630.04
is_engine_running	boolean	0	True
is_moving	boolean	0	False
latitude	double	0	-1.601798
longitude	double	0	29.883936
speed_kmh	double	0	1.7

maintenance_logs

140 rows · FK vehicle_id 

Column	Type	Nulls	Sample
maintenance_id	string	0	MTN_8B60353E
vehicle_id	string	0	VEH_000001
maintenance_type	string	0	Transmission Service
service_date	timestamp	0	2025-08-24 07:46:48
next_maintenance_date	timestamp	0	2025-10-27 07:46:48
cost	double	0	1759.43
duration_hours	double	0	1.36
technician_name	string	0	Jeffrey Gardner
service_center	string	0	Shannon LLC
description	string	0	Act guy program skill stop animal.

parts_replaced	string	0	Engine oil, Filter
tire_pressure_front_kpa	double	0	648.95
tire_pressure_rear_kpa	double	0	621.0
breakdown_recorded	boolean	0	False
breakdown_description	string	90 (64%)	set only when breakdown_recorded=True
status	string	0	Scheduled

PART 3

agl_driver_data_output — Driver Monitoring

All 6 tables exported in CSV, JSON, Parquet and XML.

KEY FINDING

Every table in this folder uses a **4-digit** `vehicle_id` format (`VEH_0001`) generated independently from the fleet domain's **6-digit** format (`VEH_000001`) — confirmed zero overlap. `driver_id` , however, is consistent everywhere in this folder (200/200 match the `drivers` table). Full detail in Part 8.

drivers

200 rows · primary key: `driver_id` (`DRV_000001`)

Column	Type	Nulls	Sample
<code>driver_id</code>	string	0	DRV_000001
<code>first_name</code>	string	0	Debbie
<code>last_name</code>	string	0	Sanchez
<code>email</code>	string	0	cynthia39@example.org
<code>phone</code>	string	0	001-502-814-5104x29250
<code>license_number</code>	string	0	DL3191854
<code>license_class</code>	string	0	F
<code>license_expiry</code>	timestamp	0	2026-12-06 17:28:40
<code>date_of_birth</code>	timestamp	0	2005-06-06 17:28:40
<code>hire_date</code>	timestamp	0	2018-02-20 17:28:40
<code>status</code>	string	0	On Leave
<code>vehicle_type</code>	string	0	Trailer
<code>home_terminal</code>	string	0	Gatesshire
<code>emergency_contact</code>	string	0	Lisa Green
<code>emergency_phone</code>	string	0	+1-506-442-4155x25296
<code>total_experience_years</code>	double	0	7.5
<code>last_medical_exam</code>	timestamp	0	2026-05-23 17:28:40
<code>next_medical_exam</code>	timestamp	0	2027-06-27 17:28:40
<code>preferred_language</code>	string	0	English
<code>rating</code>	double	0	4.39
<code>total_missions</code>	integer	0	26
<code>total_km_driven</code>	double	0	24600.64

behavioral_events

10,000 rows · FK `driver_id`  (200/200) · FK `vehicle_id`  4-digit format, no overlap with fleet `vehicles`

29 columns total; most of the columns below are **event-type-specific** and heavily null by design (a "Harsh Braking" event populates `deceleration_rate` / `distance_to_obstacle_m`; a "Speeding" event populates `excess_speed_kmh`; a "Seatbelt" event populates `duration_unbuckled_seconds`, etc.) — this is sparse-by-design, not missing data.

Column	Type	Nulls	Sample
event_id	string	0	EVT_D6A39E29
driver_id	string	0	DRV_000066
event_type	string	0	Harsh Braking
timestamp	timestamp	0	2026-07-10 04:09:40
severity	string	0	Critical
latitude	double	0	-1.607007
longitude	double	0	29.592464
speed_kmh	double	0	65.21
acceleration_g	double	0	-0.929
brake_pressure_percent	double	0	26.85
steering_angle_deg	double	0	15.63
engine_rpm	integer	0	2034
gear_position	integer	0	0
vehicle_id	string	0	VEH_0005
weather_condition	string	0	Rain
road_condition	string	0	Fair
traffic_density	string	0	Gridlock
deceleration_rate	double	7,467 (75%)	Harsh-braking events only
distance_to_obstacle_m	double	7,467 (75%)	Harsh-braking events only
acceleration_rate	double	8,429 (84%)	Rapid-acceleration events only
engine_load_percent	double	8,429 (84%)	Rapid-acceleration events only
speed_limit_kmh	double	9,210 (92%)	Speeding events only
excess_speed_kmh	double	9,210 (92%)	Speeding events only
idle_duration_minutes	double	8,063 (81%)	Excessive-idling events only
fuel_consumed_l	double	8,063 (81%)	Excessive-idling events only
duration_unbuckled_seconds	double	9,018 (90%)	Seatbelt events only
warning_count	double	9,018 (90%)	Seatbelt events only
lateral_acceleration	double	8,505 (85%)	Sharp-cornering events only
curve_radius_m	double	8,505 (85%)	Sharp-cornering events only

driver_logs

22,636 rows · FK `driver_id`  · FK `vehicle_id`  same 4-digit mismatch confirmed

Column	Type	Nulls	Sample
log_id	string	0	LOG_09502791
driver_id	string	0	DRV_000001

duty_status	string	0	Off Duty
start_time	timestamp	0	2026-07-15 06:00:00
end_time	timestamp	0	2026-07-15 06:17:00
duration_minutes	integer	0	17
vehicle_id	string	0	VEH_0036
location	string	0	Riveratown
latitude	double	0	-2.037001
longitude	double	0	30.032006
remarks	string	15,794 (70%)	free text, mostly empty
is_break	boolean	0	True
is_overtime	boolean	0	False
overtime_hours	double	0	0.0
break_required	boolean	0	False
break_taken	boolean	19,774 (87%)	only set when break_required=True
violation_noted	boolean	2,308 (10%)	—

app_telematics

623 rows · FK driver_id ✓ · contains the only **nested (non-flat)** column found in the whole dataset

Column	Type	Nulls	Sample
session_id	string	0	APP_E2BA1711
driver_id	string	0	DRV_000002
login_time	timestamp	0	2026-07-13 16:33:40
logout_time	timestamp	0	2026-07-13 22:13:12
app_active_time_seconds	integer	0	4265
app_state	string	0	Active
device_id	string	0	DEV_7772dc
device_type	string	0	Android
app_version	string	0	5.9.5
screen_interactions	integer	0	28
screen_state	string	0	Locked
data_sync_status	string	0	Partial
offline_data_size_mb	double	0	21.44
sync_delay_seconds	integer	0	116
battery_level_at_login	double	0	20.18
network_type	string	0	Offline
signal_strength	string	0	Excellent
last_known_location	struct<lat: double, lon: double> (nested)	0	{'lat': -1.890936, 'lon': 29.909521}

NOTE

In the CSV/XML exports this struct is flattened to a JSON-looking string; in the Parquet export (read directly here) it is a true nested struct with two sub-fields. Plan for this explicitly in ingestion — it's the one column in all 31 tables that isn't flat.

fatigue_data

829 rows · FK driver_id ✓ · columns are sparse by fatigue_type (Yawning / Drowsiness / Distraction / etc.)

Column	Type	Nulls	Sample
fatigue_id	string	0	FAT_F79C6491
driver_id	string	0	DRV_000002
fatigue_type	string	0	Yawning
timestamp	timestamp	0	2026-06-16 10:14:41
severity	string	0	Medium
duration_seconds	integer	0	28
latitude	double	0	-1.554639
longitude	double	0	29.74717
alert_triggered	boolean	0	False
response_time_seconds	double	564 (68%)	only when alert_triggered=True
driver_acknowledged	boolean	573 (69%)	only when alert_triggered=True
frequency_per_minute	double	608 (73%)	Yawning events
eye_closure_percent	double	608 (73%)	Yawning/Drowsiness events
heart_rate_bpm	integer	0	78
skin_temperature_c	double	0	36.31
galvanic_skin_response	double	0	9.07
head_position	string	0	Nodding
eye_speed	double	0	423.46
distraction_source	string	714 (86%)	Distraction events only
phone_type	string	701 (85%)	Phone-distraction events only

pre_trip_inspections

519 rows · FK driver_id ✓ · FK vehicle_id ✗ same 4-digit mismatch

Column	Type	Nulls	Sample
inspection_id	string	0	INS_2A1AB61D
driver_id	string	0	DRV_000002
vehicle_id	string	0	VEH_0027
inspection_date	timestamp	0	2026-07-07 22:36:41
inspection_type	string	0	Pre-Trip
inspection_items	string (embedded JSON array)	0	[{"item":"Lights","status":"Pass","notes":null}, ...]
pass_count	integer	0	6
fail_count	integer	0	1

defect_description	string	414 (80%)	only when fail_count > 0
action_taken	string	409 (79%)	only when fail_count > 0
status	string	0	Pending Review
location	string	0	Jonesmouth
latitude	double	0	-2.034966
longitude	double	0	30.315851
weather_condition	string	0	Rain
temperature_c	double	0	14.39

PART 4

agl_coldchain_data_output — Cold-Chain

All 6 tables exported in CSV, JSON, Parquet and XML.

containers

100 rows · primary key: container_id (CONT_000001)

Column	Type	Nulls	Sample
container_id	string	0	CONT_000001
container_type	string	0	20ft
product_type	string	0	Pharmaceuticals
setpoint_temp_c	double	0	6.4
temp_min_c	double	0	2.0
temp_max_c	double	0	8.0
setpoint_humidity_percent	double	0	37.3
humidity_min_percent	double	0	30.0
humidity_max_percent	double	0	60.0
capacity_m3	double	0	33.2
current_fill_percent	double	0	21.9
manufacturer	string	0	Siemens
model	string	0	R792-C
year_manufactured	integer	0	2023
last_maintenance_date	timestamp	0	2026-07-03 17:24:34
next_maintenance_date	timestamp	0	2026-09-29 17:24:34
status	string	0	Idle
assigned_warehouse	string — warehouse NAME, not warehouse_id	30 (30%)	Kigali SEZ Cold Storage 3
gps_latitude	double	0	-2.134413
gps_longitude	double	0	29.732661

warehouses

10 rows · primary key: warehouse_id (WH_001 ... WH_010)

Column	Type	Nulls	Sample
warehouse_id	string	0	WH_001
warehouse_name	string	0	Kigali SEZ Cold Storage 1
address	string (multi-line)	0	41944 Johnson Mall / Natashaland, VA 78435
total_area_m2	double	0	1163.07

cooling_capacity_kw	double	0	2243.62
number_of_bays	integer	0	12
product_types	string (comma-separated list)	0	Meat & Seafood,Dairy Products,Chemicals
temperature_zone	string	0	Ambient (15°C to 25°C)
power_backup_hours	double	0	21.46
generator_capacity_kva	double	0	937.42
manager_name	string	0	Emily Doyle
contact_phone	string	0	(641)572-6095
contact_email	string	0	jonesalyssa@example.net
operating_hours	string	0	06:00-22:00
certification	string	0	ISO 22000
status	string	0	Temporary Closure

sensors

246 rows · primary key: `sensor_id` · FK `container_id`  (37 distinct containers referenced)

Column	Type	Nulls	Sample
sensor_id	string	0	SEN_000001
sensor_type	string	0	Humidity Sensor
manufacturer	string	0	Siemens
model	string	0	M435-HU
container_id	string	62 (25%)	CONT_000007 — null when sensor sits in a warehouse instead
warehouse_id	string — warehouse NAME, misnamed column	59 (24%)	Kigali Pharma Storage
location	string	0	Door
battery_level_percent	double	0	37.6
signal_strength_dbm	double	0	-49.87
calibration_date	timestamp	0	2026-03-28 17:24:34
next_calibration_date	timestamp	0	2026-08-06 17:24:34
calibration_interval_days	integer	0	180
status	string	0	Maintenance
firmware_version	string	0	v4.0.5
last_reading	always null (100%)	246 / 246	—
last_reading_timestamp	always null (100%)	246 / 246	—

DATA QUALITY

`last_reading` and `last_reading_timestamp` are 100% empty across all 246 rows — the generator never populated them (likely intended to be back-filled from `sensor_readings` but never was). Treat these two columns as placeholders, not usable fields, unless you backfill them yourself.

sensor_readings

7,529 rows · FK `sensor_id` · FK `container_id` (nullable,  28/28 match when present) · FK `warehouse_id` (nullable, holds warehouse **names**, not codes — 8/11 distinct values match a warehouse_name)

Column	Type	Nulls	Sample
reading_id	string	0	READ_B8AE2977
sensor_id	string	0	SEN_000002
container_id	string	1,299 (17%)	CONT_000007
warehouse_id	string	1,715 (23%)	Kigali Pharma Storage
timestamp	timestamp	0	2026-07-06 17:24:34
temperature_c	double	0	6.14
humidity_percent	double	0	99.17
pressure_hpa	double	0	1014.3
battery_level_percent	double	0	18.7
signal_strength_dbm	double	0	-43.7
is_anomaly	boolean	0	False
sensor_value	double	0	80.41

4,515 of 7,529 rows (60%) have **both** container_id and warehouse_id populated (a sensor inside a container that is currently sitting in a warehouse bay); the remainder have exactly one populated. No row has both null.

reefer_status

2,086 rows · FK `container_id`  (37/37 match, all reefer-equipped containers)

Column	Type	Nulls	Sample
status_id	string	0	RS_B1B85635
container_id	string	0	CONT_000007
timestamp	timestamp	0	2026-07-14 17:24:34
power_status	string	0	On
compressor_cycles_per_hour	integer	0	3
defrost_cycle_count	integer	0	0
fuel_level_percent	double	0	28.87
battery_level_percent	double	0	89.23
cooling_unit_temp_c	double	0	9.35
condenser_pressure_bar	double	0	25.91
evaporator_pressure_bar	double	0	5.2
fan_speed_rpm	integer	0	618
runtime_hours	double	0	7103.46

sensor_health

246 rows · 1:1 with `sensors` · same container_id/warehouse_id-as-name pattern

Column	Type	Nulls	Sample
health_id	string	0	HLTH_791726C0
sensor_id	string	0	SEN_000001
container_id	string	62 (25%)	CONT_000007
warehouse_id	string (name)	59 (24%)	Kigali Pharma Storage
check_timestamp	timestamp	0	2026-07-08 17:24:34
battery_level_percent	double	0	37.6
signal_strength_dbm	double	0	-49.87
calibration_status	string	0	Valid
firmware_version	string	0	v4.0.5
data_transmission_rate	double	0	97.51
packet_loss_percent	double	0	3.79
response_time_ms	double	0	207.27
error_code	string	0 (value "None" as text)	None
maintenance_recommended	boolean	0	False
health_score	integer	0	73

PART 5

agl_delivery_data_output — Delivery & Milestones

All 5 tables exported in CSV, JSON, Parquet and XML.

KEY FINDING

Every table in this folder keys off a `shipment_id` in the format `AGL-#####` — completely different from `agl_shipment_data.shipment_id` 's `SHIP-#####` format. Both sides have exactly 1,000 unique shipments, but **zero** of the IDs match. Full detail in Part 8.

milestones

10,000 rows = exactly 10 rows × 1,000 shipments · 10 distinct `milestone_name` values, ordered 1–10: Booking Confirmed → Cargo Received → Departed Origin → Arrived at Border → Customs Cleared → Border Cleared → Departed Border → Arrived at Destination → Out for Delivery → Delivered

Column	Type	Nulls	Sample
shipment_id	string	0	AGL-00000001
milestone_name	string	0	Booking Confirmed
milestone_order	integer (1–10)	0	1
timestamp	timestamp	0	2026-04-25 17:33:30
completed	boolean	0	True
location	string	0	Gisenyi
notes	string	6,994 (70%)	—
border_post	string	7,000 (70%)	only set on border-related milestones

exceptions

587 rows · sparse by `exception_type` (10 roughly even types: Vehicle Breakdown 66, Over 62, Short 60, Documentation Issue 60, Package Lost 59, Traffic Delay 58, Weather Delay 57, Damaged 56, Missed Connection 56, Customs Hold 53) — each type populates a different subset of the OS&D/delay columns below

Column	Type	Nulls	Sample
exception_id	string	0	EXC_E20BE39A
shipment_id	string	0	AGL-00000005
exception_type	string	0	Weather Delay
occurred_at	timestamp	0	2026-04-19 23:10:56
severity	string	0	Low
description	string	0	Minute stage tonight smile.
action_taken	string	205 (35%)	—
resolution_date	timestamp	587 (100%)	always null in the sample checked
status	string	0	Resolved
assigned_to	string	309 (53%)	Robert Black

delay_reason	string	359 (61%)	Traffic/Weather/Breakdown types
delay_duration_hours	double	359 (61%)	Traffic/Weather/Breakdown types
damage_category	string	531 (90%)	Damaged type only
damage_description	string	531 (90%)	Damaged type only
damage_value	double	531 (90%)	Damaged type only
damaged_items	double	531 (90%)	Damaged type only
short_quantity	double	527 (90%)	Short type only
expected_quantity	double	527 (90%)	Over/Short types only
actual_quantity	double	527 (90%)	Over/Short types only
over_quantity	double	525 (89%)	Over type only
measured_weight	double	525 (89%)	Over type only
documented_weight	double	525 (89%)	Over type only

osd_logs

564 rows — a separate Over/Short/Damaged claim log, distinct from the OS&D columns embedded in `exceptions`

Column	Type	Nulls	Sample
osd_id	string	0	OSD_72605BD7
shipment_id	string	0	AGL-00000015
osd_type	string	0	Over
item_description	string	0	Standard would impact she us.
quantity	integer	0	38
unit	string	0	boxes
reported_by	string	0	Mary Miranda
reported_date	timestamp	0	2026-06-22 17:20:09
verified_by	string	310 (55%)	Mark Jones
verified_date	timestamp	564 (100%)	always null in the sample checked
discrepancy_value	double	0	4552.66
currency	string	0	EUR
status	string	0	Claim Filed
notes	string	0	Action foot cultural likely international turn.
claim_filed	boolean	0	False
claim_amount	double	395 (70%)	only when claim_filed=True

proof_of_delivery

898 rows (not all 1,000 shipments have a POD yet — some are still in transit) · 2 embedded-JSON string columns and 2 base64-looking signature strings

Column	Type	Nulls	Sample
pod_id	string	0	POD_AGL-00000001

shipment_id	string	0	AGL-00000001
recipient_name	string	0	Duane Winters
recipient_contact	string	0	(842)399-6745
recipient_email	string	0	heather61@example.net
recipient_signature	string (base64-encoded)	0	U0IHTkVEX0JZX0RVQU5FX1dJTIRFUInfMTBiZWm1...
delivery_agent_name	string	0	Jessica Wise
delivery_agent_signature	string (base64-encoded)	0	U0IHTkVEX0JZX0pFU1NJQ0FfV0ITRV8xN2VIN2Q2...
delivery_timestamp	timestamp	0	2026-04-28 19:32:54
delivery_location	string	0	Nyamata
cargo_condition_summary	string	0	Satisfactory
cargo_condition_details	string (embedded JSON array)	0	[{"item":"Item_1","condition":"Satisfactory","remarks":"..."}]
photo_evidence	string (embedded JSON array)	0	[{"photo_id":"PHOTO_152BBBD0","description":"Cargo photo 1",...}]
temperature_at_delivery_c	double	0	18.25
humidity_at_delivery_percent	double	0	51.32
delivery_notes	string	448 (50%)	—
received_by	string	0	James Walters

delivery_metrics

1,000 rows = one computed row per shipment (in the AGL-##### namespace)

Column	Type	Nulls	Sample
total_transit_hours	double	0	73.99
total_transit_days	double	0	3.08
customs_to_delivery_hours	double	0	34.64
border_processing_hours	double	0	5.89
expected_delivery	timestamp	0	2026-04-30 17:33:30
on_time	boolean	0	True
shipment_id	string	0	AGL-00000001

PART 6

agl_external_data_output — External / Market Data

All 5 tables exported in CSV, JSON, Parquet and XML. None of these tables carry a shipment/vehicle/driver foreign key — they're purely contextual, joined by `corridor_name` + time (weather, traffic) or `country` + time (fuel, exchange rates, market).

weather_data

5,000 rows · `corridor_name` ∈ {Kigali-Gisenyi, Kigali-Rusumo, Kigali-Gatuna} — the same 3 corridors used in `trips` and `traffic_data` ✓

Column	Type	Nulls	Sample
weather_id	string	0	WTH_933C35A1
corridor_name	string	0	Kigali-Gisenyi
location_name	string	0	Rubavu
latitude	double	0	-1.91
longitude	double	0	29.98
timestamp	timestamp	0	2026-06-22 17:39:15
is_forecast	boolean	0	False
weather_condition	string	0	Clear
temperature_c	double	0	28.41
precipitation_mm	double	0	13.53
wind_speed_kmh	double	0	53.53
wind_direction_deg	double	0	31.3
humidity_percent	double	0	51.64
pressure_hpa	double	0	981.49
visibility_km	double	0	4.76
cloud_cover_percent	double	0	50.54
uv_index	double	0	6.2
weather_alert	string	mostly "None" as text	None

traffic_data

2,000 rows · same 3 corridors · event-related columns are populated only when `has_event=True` (306 of 2,000 rows, exact match with non-null `event_type` — confirmed clean sparse design)

Column	Type	Nulls	Sample
traffic_id	string	0	TRA_1B836464
corridor_name	string	0	Kigali-Rusumo
location_name	string	0	Kigali-Rusumo Corridor
latitude	double	0	-1.897121

longitude	double	0	29.945299
timestamp	timestamp	0	2026-06-16 17:39:16
congestion_index	double	0	0.74
traffic_condition	string	0	Moderate
road_condition	string	0	Good
average_speed_kmh	double	0	76.12
travel_time_estimate_minutes	double	0	14.31
road_closure	boolean	0	False
construction_delay	boolean	0	False
construction_delay_days	integer	0	0
has_event	boolean	0	False
event_type	string	1,694 (85%)	only when has_event=True
event_description	string	1,694 (85%)	only when has_event=True
event_duration_hours	double	1,694 (85%)	only when has_event=True
affected_route	string	1,032 (52%)	Kigali-Rusumo Route
alternative_route	string	1,430 (72%)	—
border_wait_time_hours	double	1,309 (65%)	only at border-adjacent readings
number_of_trucks_waiting	double	1,309 (65%)	only at border-adjacent readings
customs_clearance_time_hours	double	1,309 (65%)	only at border-adjacent readings
documentation_check_time_minutes	double	1,309 (65%)	only at border-adjacent readings

fuel_prices

192 rows · 8 countries: Rwanda, Kenya, Uganda, Tanzania, Burundi, South Sudan, DRC, Ethiopia

Column	Type	Nulls	Sample
fuel_price_id	string	0	FPR_OD47DEB0
country	string	0	Rwanda
timestamp	timestamp	0	2026-01-28 17:39:16
fuel_price_index	double	0	98.9
fuel_price_usd	double	0	0.0824
week_number	integer	0	1
price_trend	string	0	Downward
supply_status	string	0	Tight

exchange_rates

1,440 rows · same 8 countries/currencies

Column	Type	Nulls	Sample
exchange_rate_id	string	0	EXR_9DB90981
country	string	0	Rwanda

currency	string	0	RWF
timestamp	timestamp	0	2026-01-16 17:39:16
exchange_rate_to_usd	double	0	1203.56
exchange_rate_to_rwf	double	0	1.003
day_of_year	integer	0	1
rate_change_percent	double	0	0.3
market_sentiment	string	0	Neutral

market_data

1,000 rows · same 8 countries · holiday fields populated on only 44 of 1,000 rows (the rest are non-holiday days)

Column	Type	Nulls	Sample
market_id	string	0	MKT_4ACD9336
country	string	0	Burundi
currency	string	0	BIF
timestamp	timestamp	0	2026-06-22 17:39:16
fuel_price_index	double	0	93.57
fuel_price_usd	double	0	0.0332
exchange_rate_to_usd	double	0	2820.88
exchange_rate_trend	string	0	Depreciating
inflation_rate_percent	double	0	13.53
interest_rate_percent	double	0	9.01
gdp_growth_rate_percent	double	0	2.32
trade_balance	string	0	Deficit
import_duty_rate_percent	double	0	12.74
export_volume_index	double	0	91.63
is_holiday	boolean	0	False
holiday_name	string	956 (96%)	only when is_holiday=True
holiday_type	string	956 (96%)	only when is_holiday=True
holiday_date	string	956 (96%)	only when is_holiday=True
market_volatility	string	0	High
business_confidence_index	double	0	63.23
consumer_spending_index	double	0	80.36
logistics_performance_index	double	0	3.7

PART 7

Format Availability & Corrected Row Counts

Folder	Table	Verified rows	CSV	JSON	Parquet	XML
agl_data_output	agl_shipment_data	1,000	✓	✓	✓	✓
	customers_master	200	✓	✓	—	—
	shippers_master	100	✓	✓	—	—
	consignees_master	150	✓	✓	—	—
agl_fleet_data_output	vehicles	50	✓	✓	✓	✓
	trips	500	✓	✓	✓	✓
	gps_tracking	201	✓	✓	✓	✓
	telematics	182	✓	✓	✓	✓
	maintenance_logs	140	✓	✓	✓	✓
agl_driver_data_output	drivers	200	✓	✓	✓	✓
	behavioral_events	10,000	✓	✓	✓	✓
	driver_logs	22,636	✓	✓	✓	✓
	app_telematics	623	✓	✓	✓	✓
	fatigue_data	829	✓	✓	✓	✓
	pre_trip_inspections	519	✓	✓	✓	✓
agl_coldchain_data_output	containers	100	✓	✓	✓	✓
	warehouses	10	✓	✓	✓	✓
	sensors	246	✓	✓	✓	✓
	sensor_readings	7,529	✓	✓	✓	✓
	reefer_status	2,086	✓	✓	✓	✓
	sensor_health	246	✓	✓	✓	✓
agl_delivery_data_output	milestones	10,000	✓	✓	✓	✓
	exceptions	587	✓	✓	✓	✓
	osd_logs	564	✓	✓	✓	✓
	proof_of_delivery	898	✓	✓	✓	✓
	delivery_metrics	1,000	✓	✓	✓	✓
agl_external_data_output	weather_data	5,000	✓	✓	✓	✓
	traffic_data	2,000	✓	✓	✓	✓
	fuel_prices	192	✓	✓	✓	✓
	exchange_rates	1,440	✓	✓	✓	✓
	market_data	1,000	✓	✓	✓	✓

NOTE

Only the 3 master tables in `ag1_data_output` are missing Parquet/XML. Every other table across all 6 folders genuinely has all 4 formats — the earlier document's mistake was assuming that applied universally, when it was really just those 3.

PART 8

Cross-Domain Key Integration Findings

This is the section that explains why joins across domains may have produced empty or all-NULL results. Every row below was verified by actually computing the overlap between two ID sets — not assumed from column names.

Relationship	Status	What was actually checked
agl_shipment_data.customer_id ↔ customers_master.customer_id	✓ Clean	199/199 unique shipment customer_ids found in the master table
agl_shipment_data.shipper_id ↔ shippers_master.shipper_id	✓ Clean	100/100 match
agl_shipment_data.consignee_id ↔ consignees_master.consignee_id	✓ Clean	150/150 match
fleet_vehicles.vehicle_id ↔ trips.vehicle_id / gps_tracking.vehicle_id	✓ Clean	50/50 and 5/5 match, both use <code>VEH_000001</code> (6-digit)
coldchain_containers.container_id ↔ sensors / sensor_readings / reefer_status.container_id	✓ Clean	37/37, 28/28, 37/37 match
drivers.driver_id ↔ behavioral_events / driver_logs / fatigue_data / app_telematics / pre_trip_inspections.driver_id	✓ Clean	200/200 match on behavioral_events (spot-checked)
weather_data / traffic_data / trips — corridor_name	✓ Clean	Same 3 values everywhere: Kigali-Gisenyi, Kigali-Rusumo, Kigali-Gatuna
fuel_prices / exchange_rates / market_data — country	✓ Clean	Same 8 countries in all three
agl_shipment_data.shipment_id ↔ milestones / exceptions / osd_logs / proof_of_delivery / delivery_metrics.shipment_id	✗ Zero overlap	Shipment domain uses <code>SHIP_#####</code> (1,000 unique); delivery domain uses <code>AGL-#####</code> (1,000 unique in each table). Same count, same grain, completely disjoint ID space — these look like two independent generator runs that were never keyed to the same IDs.
fleet_vehicles.vehicle_id ↔ driver-domain vehicle_id (behavioral_events, driver_logs, pre_trip_inspections)	✗ Zero overlap	Fleet uses <code>VEH_000001</code> (6-digit); every driver-domain table uses <code>VEH_0001</code> (4-digit). Confirmed on all 3 driver tables that carry a vehicle_id.
fleet_trips.trip_id ↔ fleet_gps_tracking.trip_id	✗ Zero overlap	500 unique trip_ids in <code>trips</code> , 199 unique in <code>gps_tracking</code> , no shared values — GPS pings cannot currently be attributed to a specific row in <code>trips</code> .
vehicles.assigned_driver	⚠ Name, not ID	Holds a full name string ("Paige Coleman"), not a <code>driver_id</code> — joining to <code>drivers</code> requires matching on concatenated first/last name, which is fragile (duplicate names, formatting differences).
trips.driver_name	⚠ Name, not ID	Same issue — free-text name, no <code>driver_id</code> column on <code>trips</code> at all.
containers.assigned_warehouse	⚠ Name, correctly so	Holds the warehouse's <code>warehouse_name</code> (10/10 match) — usable, but must join on <code>warehouses.warehouse_name</code> , not <code>warehouse_id</code> .

sensors / sensor_readings / sensor_health.warehouse_id	 Misnamed column	Despite the column name, it holds warehouse names (8-9 of ~11-12 distinct values match warehouse_name), not warehouse_id codes. Join on warehouses.warehouse_name .
shippers_master.shipper_id format	 Naming collision risk	Uses the same SHIP_ prefix as agl_shipment_data.shipment_id , but with 6-digit padding (SHIP_000084) vs. the shipment table's 8-digit padding (SHIP_00000144). Easy to visually confuse in code reviews — not a join failure, but a readability trap.

PRACTICAL TAKEAWAY

Before building any cross-domain join (trip↔GPS, vehicle↔driver-behavior, shipment↔delivery-event), decide how to bridge the mismatched key: either regenerate the data with shared keys, add an explicit bridge/mapping table, or normalize one side's ID format at ingestion time (e.g. strip padding and re-derive a common key). Building the join first and discovering empty results afterward is what likely happened in the pipeline you already ran.

PART 9

Other Data Quality Observations

Table.Column	Observation
sensors.last_reading, sensors.last_reading_timestamp	100% null across all 246 rows — present in the schema but never populated by the generator.
exceptions.resolution_date	Null on all 587 sampled rows — every exception in the sample is still technically "unresolved" by timestamp even where <code>status = Resolved</code> . Worth a sanity check before using this column.
osd_logs.verified_date	Same pattern — null on all 564 rows checked.
app_telematics.last_known_location	The only genuinely nested (struct) column found across all 31 tables; every other table is flat. Handle it explicitly rather than assuming uniform flat-JSON handling across the whole driver domain.
behavioral_events (29 cols), exceptions (22 cols), fatigue_data (20 cols), traffic_data (24 cols)	All four are "wide sparse" tables — most of their nullable columns are populated only for specific type/category values (event_type, exception_type, fatigue_type, has_event). This is intentional generator design, not a data defect, but it means a naive "drop columns with >50% nulls" cleaning rule would delete real, meaningful fields.
agl_shipment_data.actual_delivery_date	22% null — expected, these are shipments still in transit at generation time (no <code>status = Delivered</code> yet).
Embedded multi-line address fields	customers_master, shippers_master, consignees_master and warehouses all have an <code>address</code> column containing an embedded newline (street line \n city/state/zip line). Any tool that naively splits CSV by newline instead of respecting quoted fields will miscount rows — this is what inflated the row counts in the earlier document.
proof_of_delivery row count (898) vs agl_shipment_data (1,000)	Not every shipment has a POD yet — consistent with ~22% of shipments still lacking an <code>actual_delivery_date</code> .

AGL Rwanda Data Dictionary — Verified Structure Reference. Built by direct inspection of the Parquet/CSV files in agl_data_output, agl_fleet_data_output, agl_driver_data_output, agl_coldchain_data_output, agl_delivery_data_output and agl_external_data_output.